KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM agents WHEN to act directly, reflect, or consult knowledge

- Three situational thinking modes: Fast · Slow · Knowledgeable
- 84.33% on ALFWorld with Llama-8B — using only 15% knowledge
- arXiv: 2504.03553

ML Research Presentation

2025

The 'Flood Irrigation' Problem in Agent Training

Indiscriminate knowledge injection — regardless of situational need

1 — Trajectory Imitation

e.g. ReAct

Copies gold trajectories step-by-step without situational judgment

2 — Trial-and-Error Refinement

e.g. Reflexion

Applies external feedback indiscriminately at every decision point

3 — Knowledge-Augmented Planning

e.g. KnowAgent, WKM

Injects domain knowledge at every step regardless of actual need

⚠️ **All three treat knowledge as a constant input — ignoring whether the agent actually needs it at each step**

Humans use situational self-awareness

Metacognition allows humans to judge: Can I handle this directly? Do I need to pause and reflect? Or must I consult external resources? Current agents lack this metacognitive layer entirely.

Three Situational Modes: Fast · Slow · Knowledgeable Thinking

KnowSelf classifies each decision step before acting

Agent at decision step

Can reflection fix the error?

gent produce correct action directly?

YES ▶

⚡ FAST THINKING

Direct action generation
• No reflection needed
• Minimal token overhead
• Token: <fast>

YES ▼

NO ▼

🔄 SLOW THINKING

Self-reflection loop
• Initial action was wrong
• Reflection corrects it
• Token: <reflect>

📚 KNOWLEDGEABLE THINKING

External knowledge retrieval
• Reflection insufficient
• Consults knowledge base
• Token: <knowledge>

Key insight: Special tokens (<fast> / <reflect> / <knowledge>) are inserted during training — the model learns to self-classify its own cognitive state at each step via heuristic labeling.

KnowSelf Pipeline: Data Construction → Two-Stage Training → Self-Aware Inference

Three-step framework for teaching agents metacognitive self-regulation

Step 1

Self-Awareness Data Construction

- Agent self-explores environments
- Each step evaluated against gold action
- Fast: predicted = gold action
- Slow: reflection corrects prediction
- Know: reflection still incorrect
- Special tokens inserted into trajectories

Step 2

Two-Stage Training

- Stage A — Supervised Fine-Tuning (SFT)
  Teaches initial self-awareness patterns from heuristically labeled trajectories

- Stage B — Reward-based Policy Optimization (RPO)
  Refines token generation quality

Step 3

Self-Aware Inference

- Agent generates special token first

<fast> → direct action output
<reflect> → runs reflection loop
<knowledge> → queries external KB

- No fixed knowledge injection
- Situation-conditioned response

Data efficiency: No expensive human annotation required — heuristic labeling on self-explored trajectories is sufficient for the model to learn metacognitive self-awareness.

KnowSelf Correctly Heats a Mug and Places a Saltshaker — Where Baselines Fail

Qualitative comparison: KnowSelf vs. OpenAI-O1 and DeepSeek-R1

Task A: "Put a hot mug in cabinet"

✗ OpenAI-O1 (baseline)

Opens microwave → incorrectly attempts to take an egg from inside

✓ KnowSelf — Llama-8B [<reflect>]

Recognises the microwave is already open → heats the mug correctly → places in cabinet

Task B: "Put a saltshaker in drawer"

✗ DeepSeek-R1 (baseline)

Reflects that the drawer is open → incorrectly tries to close it first → task fails

✓ KnowSelf — Llama-8B [<knowledge>]

Applies domain knowledge: place the object directly without moving unrelated items

Key observation:

Large proprietary models (OpenAI-O1, DeepSeek-R1) apply strong but undirected reasoning — they cannot self-regulate which cognitive mode is appropriate. KnowSelf's explicit token-based situational awareness allows a smaller open model (Llama-8B) to outperform them on these planning tasks.

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

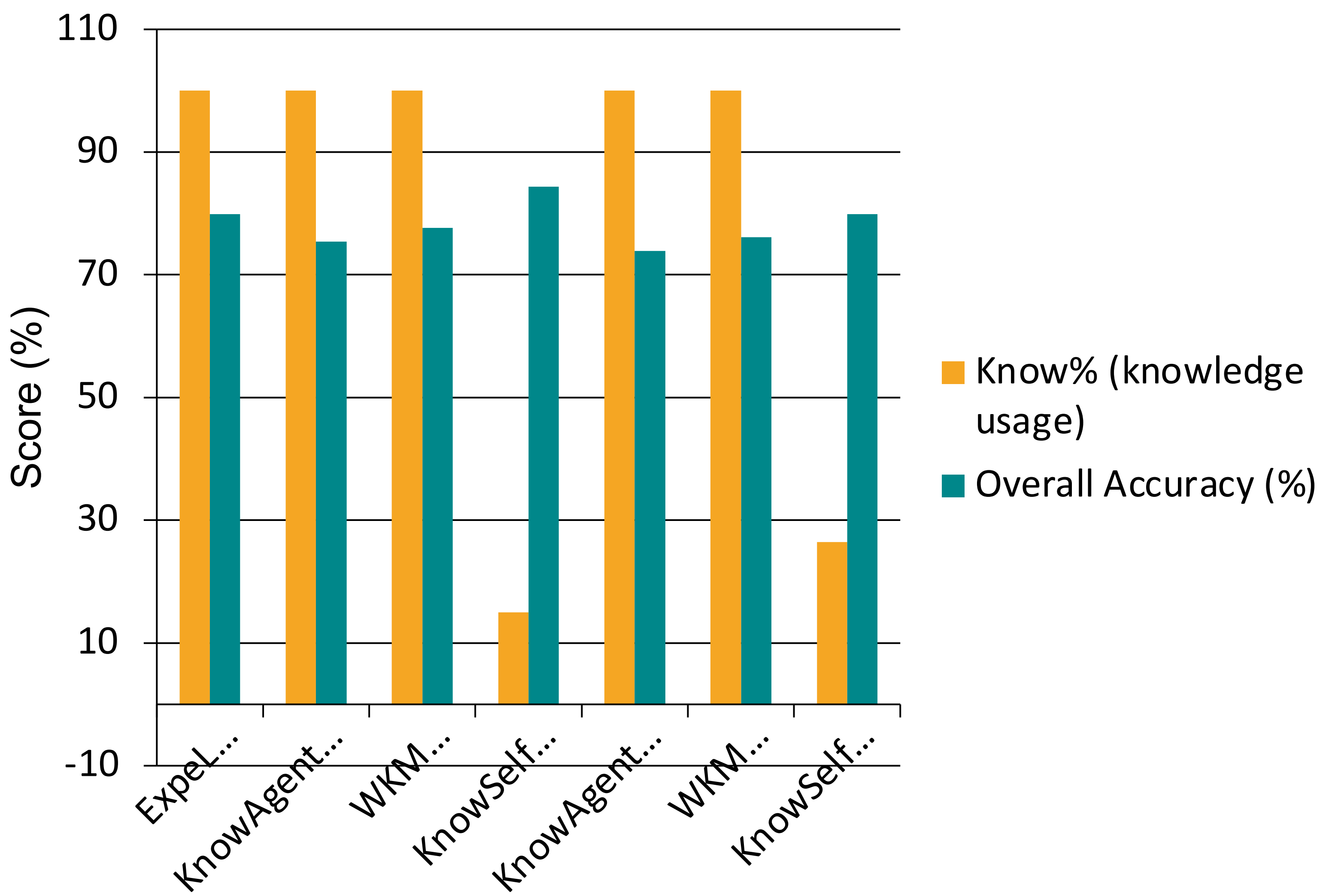
Main results: success rate (%) across six task types — Know% = proportion of steps using external knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	**KnowSelf**	**15.01%**	**91.67**	**87.10**	**91.30**	**85.71**	**77.78**	**64.71**	**84.33**
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	**KnowSelf**	**26.41%**	**87.50**	**93.55**	**73.91**	**76.19**	**83.33**	**52.94**	**79.85**

★ KnowSelf rows highlighted green. Know% = % steps using external knowledge. KnowSelf (Llama-8B) surpasses GPT-4o ExpeL (100% knowledge) with only 15% — achieving 84.33% vs 79.85%.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage

Selective knowledge use (~15–26%) outperforms 100% injection — across model scales



15.01 %

knowledge usage for KnowSelf (Llama-8B)
vs. 100% for ExpeL, KnowAgent, WKM

84.33 %

overall accuracy (Llama-8B)
Best among all Llama-8B methods

26.41 %

knowledge usage for KnowSelf (Gemma-2B)
still matches GPT-4o ExpeL at 79.85%

6.7× less knowledge

needed to match or beat full-knowledge
baselines on ALFWorld

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

KnowSelf — arXiv 2504.03553

01

Indiscriminate injection is wasteful and suboptimal

All three current paradigms (imitation, trial-and-error, knowledge-augmented) treat knowledge as a constant input — ignoring whether the agent needs it at each specific step.

02

Situational self-awareness can be learned from lightweight heuristics

KnowSelf's special-token mechanism requires no expensive human annotation — heuristic labeling of self-explored trajectories is sufficient for an LLM to learn metacognitive state classification.

03

KnowSelf (Llama-8B) beats strong baselines including GPT-4o

84.33% on ALFWorld vs. 76.12% for GPT-4o ReAct and 79.85% for GPT-4o ExpeL — a small open model surpasses proprietary systems through selective reasoning.

04

Only 15–26% knowledge usage needed vs. 100% for competing methods

KnowSelf queries the knowledge base only when genuinely needed, reducing inference cost dramatically while achieving higher accuracy — confirming selective > indiscriminate injection.

05

Generalises across model scales — Gemma-2B reaches 79.85%

Even the 2B parameter Gemma model matches GPT-4o ExpeL, demonstrating that the self-awareness paradigm is not limited to large models and scales down effectively.

Future directions: extend to multi-agent settings · richer knowledge bases · continuous self-calibration